

Di Miao

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE - Singapore

Aug 2021 - Jan 2023

Master of Science in Statistics

CAP: 3.97/5.0

Graduate Certificate in *Data Mining for Industry (Competence)* obtained as part of the Master's programme.

BEIJING NORMAL UNIVERSITY - Beijing, China

Sep 2017 - Jul 2021

Bachelor of Science in Statistics

GPA: 84/100

EXPERIENCE

DUKE-NUS MEDICAL SCHOOL - Singapore

Jan 2023 - Jan 2025

Senior Research Assistant

- Co-first author of a peer-reviewed research article, leading the quantitative analysis, figure design, and manuscript drafting; co-authored 2 additional research articles and 4 review articles, with contributions spanning the research process from initial literature screening and data analysis to manuscript drafting.
- Conducted comprehensive data analysis for clinical research projects using R and Python, applying models such as time series, logistic regression, random forests, and federated learning to identify significant relationships and trends.
- Crafted insightful visualizations using ggplot2, Matplotlib, Power BI, and QGIS for exploratory analysis and to communicate complex findings to multidisciplinary teams.

HONDA MOTOR (CHINA) TECHNOLOGY CO., LTD - Guangzhou, China

May 2025 - Present

Operations Support Specialist

- Spearheaded the development of the data platform's visualization layer, designing and building a comprehensive set of interactive dashboards with Quick BI, DataV, and JavaScript. Transforming complex business metrics into intuitive, actionable insights through advanced customization.
- Independently authored and optimized complex SQL queries to extract, process, and manage large-scale datasets from cross-departmental databases, ensuring high-quality data for all visualization projects.
- Partnered with cross-functional teams to analyze business requirements and data sources, delivering comprehensive reports that translated diverse needs into actionable specifications, supporting data-driven strategy.

SELECTED PUBLICATIONS

Li, Siqi, Di Miao, et al. (2024). "Federated Learning in Healthcare: A Benchmark Comparison of Engineering and Statistical Approaches for Structured Data Analysis". In: *Health Data Science* 4, p. 0196. doi: 10.34133/hds.0196. eprint: <https://spj.science.org/doi/pdf/10.34133/hds.0196>. URL: <https://spj.science.org/doi/abs/10.34133/hds.0196>.

Li, Siqi, Ziwen Wang, et al. (Oct. 2025). "Developing federated time-to-event scores using heterogeneous real-world survival data". In: *Computers in Biology and Medicine* 197, p. 111084. ISSN: 0010-4825. doi: 10.1016/j.combiomed.2025.111084. URL: <http://dx.doi.org/10.1016/j.combiomed.2025.111084>.

Li, Siqi, Qiming Wu, et al. (Aug. 12, 2025). "FairFML: fair federated machine learning with a case study on reducing gender disparities in cardiac arrest outcome prediction". In: *npj Health Systems* 2.1, p. 29. ISSN: 3005-1959. doi: 10.1038/s44401-025-00035-2. URL: <https://doi.org/10.1038/s44401-025-00035-2>.

Teo, Zhen Ling et al. (Feb. 2024). "Federated machine learning in healthcare: A systematic review on clinical applications and technical architecture". In: *Cell Reports Medicine* 5.2. ISSN: 2666-3791. doi: 10.1016/j.xcrm.2024.101419. URL: <https://doi.org/10.1016/j.xcrm.2024.101419>.

ADDITIONAL

- Advanced Analytics: Proficient in statistical modeling and machine learning. Independently designed and built AI agent for information retrieval and knowledge extraction.
- Programming & Databases: Python, R, SQL, SAS, Stata
- Visualization & Reporting: Power BI, Quick BI, DataV, Tableau, LaTeX, SPSS, FFmpeg
- Language: English (Business Professional), Mandarin (Native), Japanese (Conversational)
- Certifications:
 - Course on Coursera: *Science and Engineering of Climate Change, Climate Science for Everyone*
 - *Japanese-Language Proficiency Test (JLPT)* Level N2
 - *PADI Advanced Open Water Diver* (Professional Association of Diving Instructors)